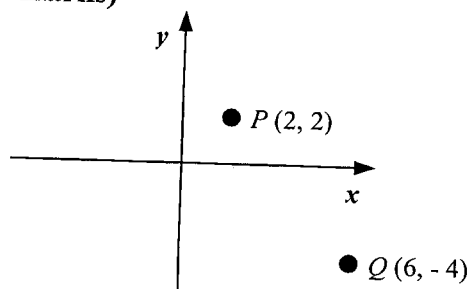


- INSTRUCTIONS:
1. Attempt all questions.
 2. Write your answers on your own paper.
 3. All necessary working must be shown.
 4. Marks will be deducted for careless or badly arranged work.

Question 1 (16 marks)

- (a) If P is the point $(2, 2)$ and Q is the point $(6, -4)$.
Find,
- (i) The gradient of the line PQ . 2
 - (ii) The equation of the line PQ 3
 - (iii) The midpoint of PQ . 2
 - (iv) The exact distance from P to Q . 3
- (b) What is the perpendicular distance of the point $(-1, -2)$ from the line $6x + 5y = 30$. 3
- (c) Find the equation of the line that has an inclination of 135° with the x axis and a y intercept of -2 . 3

Question 2 (14 marks)

- (a) find the equation of the line perpendicular to $2x + y = 6$ that passes through $(1, 3)$. 4
- (b) Find the equation of the line that passes through the point of intersection of the lines $2x - 3y + 5 = 0$ and $6x - y + 3 = 0$ and also through the point $(-2, -1)$. 5
- (c) The equations of the sides of a triangle are $2x - 3y + 5 = 0$, $2x + y = 7$ and $2x + 5y = 3$. Find the coordinates of its vertices. 5

Question 3 (12 marks)

Evaluate,

(a) $\lim_{x \rightarrow 1} \frac{x-4}{2x-8}$

(b) $\lim_{x \rightarrow 2} \frac{x^2-4}{x^2+2x-8}$

4

(c) $\lim_{x \rightarrow \infty} \left(\frac{1}{x+2} \right)$

(d) $\lim_{x \rightarrow \infty} \frac{3x^2+x-3}{2x^2-x-2}$

4

(e) (i) If $f(x) = x^2 + x$, evaluate $f(x+h) - f(x)$

2

(ii) Differentiate $y = x^2 + x$ from first principles.

2

Question 4 (15 marks)

Differentiate the following

(a) $y = x^7 + 3x^2 - 4$

(b) $y = (3x+7)^4$

4

(c) $y = \frac{x-1}{x-2}$

(d) $y = \sqrt[3]{x}$

5

(e) $y = 3x^4 + \frac{2}{x}$

(f) $y = \frac{2x^4 - 3x^3 + x}{x}$

6

Question 5 (23 marks)

Differentiate the following

(a) $y = (3x^2 - 2)(5+x)^3$

(b) $y = \frac{2}{(x^2+3)^4}$

6

(c) $y = \frac{2-x^3}{7x^2+x}$

(d) $y = 6x\sqrt{4-x^2}$

7

(e) Find the equation of the tangent and normal at $x = 3$ for the curve $y = x(2-x)$.

5

(f) Find the point on the curve $y = \sqrt{x-2}$ where the tangent is parallel to the line $2x - 2y + 3 = 0$.

5

*Mathematics - TASK 4
(2004) Yr 11.

Q1(a) P(2,2) Q(6,-4)

(i) $m = \frac{-4-2}{6-2} = \frac{-6}{4} = -\frac{3}{2}$ (2)

(ii) $y-2 = -\frac{3}{2}(x-2)$ or $2y-4 = -3x+6$ (3)
 $y = -\frac{3}{2}x + 5$ $3x+2y-10=0$

(iii) Midpoint (4,-1) (2)

(iv) $d = \sqrt{(6-2)^2 + (-4-2)^2}$
 $= \sqrt{16+36}$
 $d = \sqrt{52}$ or $= 2\sqrt{13}$ (3)

(b) $d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$
 $= \frac{|6(-1)+5(-2)+(-30)|}{\sqrt{6^2+5^2}}$
 $= \frac{|-46|}{\sqrt{61}} = \frac{46}{\sqrt{61}}$ (3)

(c) $m = \tan 135^\circ$
 $m = -1$ $b = -2$ (3)
 $\therefore y = -x - 2$ or $x+y+2=0$

Q2(a) $m_1 = -2$ $\therefore m_2 = \frac{1}{2}$ (1,3)
 $\therefore y - 3 = \frac{1}{2}(x-1)$
 $y = \frac{1}{2}x + \frac{5}{2}$
 $x - 2y + 5 = 0$ (4)

Q2(b) $2x-3y+5=0$ (1)(x3)
 $6x-y+3=0$ (2)
 $6x-9y+15=0$ (3)
 $-8y+12=0$

$y = \frac{12}{8}$
 $y = \frac{3}{2}$
 $\therefore 2x-3(\frac{3}{2})+5=0$
 $2x+\frac{5}{2}=0$
 $2x = -\frac{5}{2}$
 $x = -\frac{5}{4}$ (2,1,3)

$x = -\frac{5}{4}$ (2,1,3)
 $8(-2, -1)$

$\frac{y+1}{x+2} = \frac{\frac{1}{2}+1}{-\frac{5}{4}+2}$
 $\frac{y+1}{x+2} = \frac{\frac{3}{2}}{\frac{3}{4}}$
 $y+1 = \frac{10}{7}(x+2)$
 $7y+7 = 10x+20$
 $10x-7y+13=0$

OR (-2,-1) K-METHOD (5)
 $2x-3y+5 + k(6x-y+3) = 0$
 $2(-2)-3(-1)+5 + k(6(-2)-(-1)+3) = 0$
 $4+k(-8) = 0$
 $k = \frac{1}{2}$

$2x-3y+5 + \frac{1}{2}(6x-y+3) = 0$
 $4x-6y+10 + 6x-y+3 = 0$
 $10x-7y+13=0$

Q2(b) $2x-3y+5=0$ (1)
 $2x+y-7=0$ (2)
 $2x+5y-3=0$ (3)
 $-4y+12=0$
 $y = 3$
 $2x-3(3)+5=0$
 $2x-4=0$
 $x = 2$

(2,3)
 $4y+4=0$
 $y = -1$
 $2x+(-1)-7=0$
 $2x-8=0$
 $x = 4$

(4,-1)
 $8y-8=0$
 $y = 1$
 $2x+5(1)-3=0$
 $2x+2=0$
 $x = -1$
 $(-1,1)$ (5)

Q3

$$a) \frac{1-4}{2(1)-6} = \frac{-3}{-6} = \frac{1}{2} \quad (2)$$

$$b) \lim_{x \rightarrow 2} \frac{(x+2)(x-2)}{(x+4)(x-2)}$$

$$= \frac{2+2}{2+4} = \frac{4}{6} = \frac{2}{3} \quad (2)$$

$$c) \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1 + \frac{1}{x^2}} = \frac{0}{1} = 0 \quad (2)$$

$$d) \lim_{x \rightarrow 0} \frac{3 + \frac{1}{x} + \frac{3}{x^2}}{2 - \frac{1}{x} - \frac{2}{x^2}} = \frac{3}{2} \quad (2)$$

$$e) i) f(x+h) = (x+h)^2 + (x+h)$$

$$= x^2 + 2xh + h^2 + x + h$$

$$f(x) = x^2 + x$$

$$\therefore f(x+h) - f(x) = 2xh + h^2 + h$$

$$= h(2x+h+1) \quad (2)$$

$$(ii) \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x+h+1)}{h}$$

$$= 2x+1 \quad (2)$$

(2)

$$\frac{dy}{dx} = 2x+1$$

Q4

$$a) y' = 7x^4 \quad (2)$$

$$b) y' = 12(3x+7)^3 \quad (2)$$

$$c) y' = \frac{(x-2) \cdot 1 - (x-1) \cdot 1}{(x-2)^2}$$

$$= \frac{-1}{(x-2)^2} \quad (3)$$

$$d) y' = \frac{1}{3} x^{-\frac{2}{3}} = \frac{1}{3\sqrt[3]{x^2}} \quad (2)$$

$$e) y' = 12x^3 - 2x^2$$

$$= 12x^3 - \frac{2}{x^2} \quad (3)$$

$$f) y = 2x^3 - 3x^2 + 1 \quad (3)$$

$$y' = 6x^2 - 6x$$

Q5

$$a) y' = (3x^2-2) \cdot 3(5x)^2 + (5x)^3 \cdot 6x$$

$$y' = 3(5x)^2 [3x^2-2] + 2(5x)^3$$

$$y' = 3(5x)^2 (5x^2+10x-2) \quad (3)$$

$$b) y = 2(x^2+3)^{-4}$$

$$y' = -8(x^2+3)^{-5} \cdot 2x$$

$$= \frac{-16x}{(x^2+3)^5} \quad (3)$$

$$c) y' = \frac{(7x^2+x) \cdot (-3x^2-2) - (x^2-2) \cdot (14x+1)}{(7x^2+x)^2}$$

$$y' = \frac{-21x^4 - 3x^3 - (28x+2-14x^2-x^2)}{(7x^2+x)^2}$$

$$y' = \frac{-7x^4 - 2x^3 - 28x - 2}{(7x^2+x)^2} \quad (3)$$

Q5

$$d) y' = uv' + v$$

$$y' = 6x \cdot \frac{1}{2}(4-x^2)^{-\frac{1}{2}} \cdot (-2x)$$

$$+ (4-x^2)^{\frac{1}{2}} \cdot 6$$

$$= -6x^2(4-x^2)^{-\frac{1}{2}} + 6(4-x^2)^{\frac{1}{2}}$$

$$= 6(4-x^2)^{-\frac{1}{2}} [-x^2 + (4-x^2)]$$

$$= \frac{6}{\sqrt{4-x^2}} \cdot (4-2x^2)$$

$$= \frac{24-12x^2}{\sqrt{4-x^2}} \quad (4)$$

$$(e) y = x(2-x)$$

$$y = -x^2 + 2x$$

$$\frac{dy}{dx} = -2x + 2$$

when $x=3$ Tangent

$$\frac{dy}{dx} = -2(3) + 2$$

$$m_1 = -4$$

$$\therefore (3,3), m=-4$$

$$y+3 = -4(x-3)$$

$$y = -4x + 9$$

(5)

$$\text{when } x=3$$

$$y = 3(2-3)$$

$$y = -3$$

Normal

$$m_2 = \frac{1}{4}$$

$$(3,-3), m=\frac{1}{4}$$

$$y+3 = \frac{1}{4}(x-3)$$

$$y = \frac{1}{4}x + 2\frac{3}{4}$$

$$(f) y = (x-2)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(x-2)^{-\frac{1}{2}} \cdot 1$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x-2}}$$

$\therefore \text{when } \frac{dy}{dx} = 1$

$$1 = \frac{1}{2\sqrt{x-2}}$$

$$\sqrt{x-2} = \frac{1}{2}$$

$$x-2 = \frac{1}{4}$$

$$x = 2\frac{1}{4}$$

$$\therefore y = (2\frac{1}{4}-2)^{\frac{1}{2}}$$

$$y = \sqrt{\frac{1}{4}}$$

$$y = \frac{1}{2}$$

$$\therefore (2\frac{1}{4}, \frac{1}{2}) \quad (5)$$

$$2x-2y+3=0$$

$$y = x + \frac{3}{2}$$

$$\therefore m=1$$